

ECONOMETRICS I

Problem Set 5

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Some binary variable models

1. **CROSS SECTIONAL DATA** Answer the following using the stata datafile http://www.stata.com/data/s4poe5/data/cps5_small.dta
 - (a) How do average salaries differ between those with and without high-school education? Answer after creating a high-school dummy indicating more than 9 years of education.
 - (b) How do wages increase per year of education?
 - (c) Does the wage increase per year depend on whether high school was visited?
 - (d) How does the answer to (a) change if you distinguish between blacks and whites?
 - (e) How does the chance of having high-school education depend on being black?
 - (f) How does that chance interact with being female?
2. **TIME SERIES DATA.** Answer the following using the stata datafile **passengers.txt**.
 - (a) Plot the number of passengers vs. time.
 - (b) Estimate a liner time trend of $\log(\text{passengers})$ and plot it together with $\log(\text{passengers})$ against time.
 - (c) Generate dummies for each of the 12 months. Fit them to the trend residuals and plot them together over time.
 - (d) Regress $\log(\text{passengers})$ to both the trend and the dummies. Use that model to make a two-year forecast and plot it.
 - (e) After looking at the regression coefficients, confirm the forecast by hand for November and December in the first forecasted year.
3. **PANEL DATA.** Answer the following using **lifeexpectancyPanel.xlsx**.

- (a) Regress life expectancy on log GDP per capita without controlling for fixed effects.
- (b) Now include fixed effects as dummies.
- (c) Now estimate the fixed and random effects models with R's plm library.
- (d) Compare the results.